Total Points: [10] [U1HW6 - W8 - Kinetic and Gravitational Potential Energy Calculations]

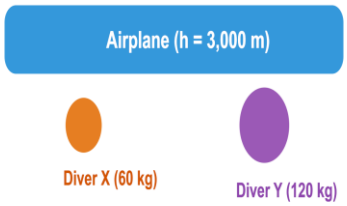
1. A 0.5 kg baseball has 100 J of kinetic energy as it crosses home plate. What is the velocity of the baseball? **[1]**

- 45 m/s
- 10 m/s
- 200 m/s
- 25 m/s
- 20 m/s
- 14.1 m/s

2. If a moving vehicle triples its velocity from 10 m/s to 30 m/s, by what factor does its kinetic energy (KE) increase? **[1]**

- It triples (3 times greater)
- It increases by 27 times
- It increases by 9 times (3 × 3)
- It increases by 6 times
- It remains completely unchanged
- It doubles (2 times greater)

3. Consider two skydivers, Diver X (m = 60 kg) and Diver Y (m = 120 kg), both standing inside an airplane ready to jump at an altitude of 3,000 meters above ground level.

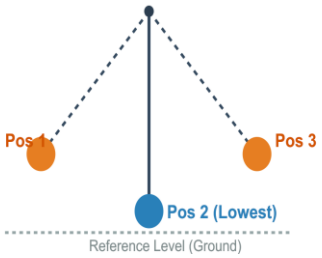


How does the gravitational potential energy (GPE) of Diver Y compare to Diver X before jumping? **[1]**

- Neither diver has GPE until they begin falling.
- Diver Y has four times the GPE of Diver X.
- Diver Y has twice (double) the GPE of Diver X because GPE is directly proportional to mass.
- Diver Y has less GPE because heavier objects experience less gravity.
- Diver Y has half the GPE of Diver X.
- Diver Y has exactly the same GPE because they are at the same height.

4. Examine the simple pendulum swinging between positions 1, 2, and 3 without air resistance:

Which statement correctly describes the energy transformation at **Position 2**? **[1]**



- GPE has increased compared to Position 1.

Total Points: [10] [U1HW6 - W8 - Kinetic and Gravitational Potential Energy Calculations]

- Both KE and GPE are equal to zero.
- The pendulum has stopped moving entirely.
- KE is at its maximum and GPE is at its minimum.
- GPE is at its maximum and IE is at its minimum.
- Total mechanical energy is doubled at Position 2.

5. A student compresses a spring launcher horizontally to propel a 0.2 kg steel ball along a smooth, level floor. If the released spring transfers 3.6 J of elastic potential energy entirely into kinetic energy of the ball, what speed does the ball attain? **[1]**

- 3.6 m/s
- 4.9 m/s
- 6.0 m/s
- 12.0 m/s
- 36.0 m/s
- 1.8 m/s

6. A 2 kg brick rests on top of a 5-meter-high scaffolding. Using g = 9.8 m/s^2, what is the gravitational potential energy (GPE) of the brick relative to the ground? **[1]**

- 20 J
- 196 J
- 10 J
- 490 J
- 49 J
- 98 J

7. A crane lifts a 250 kg steel beam straight up, giving it 24,500 J of gravitational potential energy. Taking g = 9.8 m/s^2, how high above the ground was the beam lifted? **[1]**

- 10 meters
- 8.0 meters
- 5.0 meters
- 2.5 meters
- 25 meters
- 7.5 meters

8. A remote-controlled toy race car with a mass of 1.5 kg drives down a straight track at a constant speed of 4 m/s. What is its kinetic energy (KE)? **[1]**

- 3 J
- 6 J
- 24 J
- 12 J
- 48 J
- 18.5 J

9. A 4 kg bowling ball is dropped from rest from the top of a 20-meter cliff. Neglecting air resistance and using g = 10

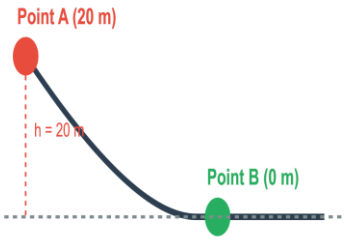
m/s^2 for simplicity, what is the velocity of the bowling ball just before it strikes the ground? **[1]**

- 5 m/s
- 40 m/s
- 20 m/s
- 400 m/s
- 10 m/s
- 200 m/s

- 156,200 J

10. An 800 kg roller coaster cart sits momentarily at rest at Point A (height = 20 m) before rolling down a frictionless track to Point B at ground level (height = 0 m).

Assuming g = 9.8 m/s^2 and zero energy loss to friction, what is the kinetic energy of the cart when it reaches Point B? **[1]**



- 312,400 J
- 0 J
- 78,100 J
- 38,400 J
- 16,000 J